A FINITE ETALE MORPHISM BECOMES A FINITE DISJOINT UNION OF ISOMORPHISMS OVER AN ETALE NEIGHBOURHOOD

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 04HJ: sources/stacks-chow/tags.txt, source lines 5870–5870. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04HL: sources/stacks-chow/tags.txt, source lines 5872–5872. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a finite étale morphism of schemes. Let $s \in S$. There exists an étale neighbourhood $(U, u) \to (S, s)$ and a finite disjoint union decomposition*

$$X_U = \coprod\nolimits_j V_j$$

of schemes such that each $V_j \to U$ is an isomorphism.

Proof. An étale morphism is unramified, hence we may apply Lemma 04HJ. As in the proof of Lemma 04HL we see that $V_{i,j} \to U$ is an open immersion and we win after replacing U by the intersection of their images. □

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